PUSHFORWARD OF INJECTIVES ALONG A FLAT QCQS MORPHISM STAYS INJECTIVE

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Editorial note. Counted span is the author's own proof environment, source lines 2906-2914, immediately following the statement at 2897-2904.

Editorial note. Hypothesis-verification pattern: flatness feeds exactness of f*, qcqs feeds quasi-coherence, the adjunction feeds the terminal lemma.

Allowed standard prerequisites (editorial).

- Quasi-coherence of pushforward. Morphisms of Spaces, Lemma spaces-morphisms-lemma-pushforward, cited at proof lines 2908-2910. f_* sends QCoh(O_X) into QCoh(O_Y) for qcqs f.
- Adjoints preserve injectives. Homology, Lemma homology-lemma-adjoint-preserve-injectives, cited at proof lines 2912-2914. Terminal input: a right adjoint whose left adjoint is exact preserves injective objects.

Source reference. Original reference 015Z: focused-seeds/geometry/S9-perf-flat-pullback-injective-quasi-coherent/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03M9: focused-seeds/geometry/S9-perf-flat-pullback-injective-quasi-coherent/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a flat, quasi-compact, and quasi-separated morphism of algebraic spaces over S. If $\mathcal{J}$ is an injective object of $QCoh(\mathcal{O}_X)$, then $f_*\mathcal{J}$ is an injective object of $QCoh(\mathcal{O}_Y)$.*

Proof. Since f is quasi-compact and quasi-separated, the functor f_* transforms quasi-coherent sheaves into quasi-coherent sheaves (Morphisms of Spaces, Lemma 03M9). The functor f^* is a left adjoint to f_* which transforms injections into injections. Hence the result follows from Homology, Lemma 015Z □

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